

And Subset Count

Time limit: 1.00 s

Memory limit: 512 MB

You are given an array of n integers. Your task is to calculate the number of non-empty subsets whose elements' *bitwise and* is equal to k for each $k = 0, 1, \dots, n$.

Input

The first line has an integer n : the size of the array.

The next line has n integers $a_1, a_2, \dots, a_n$: the contents of the array.

Output

Print $n + 1$ integers as specified above modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $0 \leq a_i \leq n$

Example

Input	Output
4 3 1 3 4	7 4 0 3 1